$S°(298.15\ K) = 237.2 \pm 13\ J·K^{-1}·mol^{-1}$

$\Delta_f H°(0\ K) = 811 \pm 17\ kJ·mol^{-1}$
$\Delta_f H°(298.15\ K) = 820 \pm 17\ kJ·mol^{-1}$

Vibrational Frequencies and Degeneracies ν, cm^{-1}
1225 (1)
63.5 (2)
2040 (1)

Ground State Quantum Weight: [1] $\sigma = 2$
Point Group: $D_{\infty h}$
Bond Distances: C–C = 1.277 Å
Bond Angle: C–C–C = 180°
Rotational Constant: $B_0 = 0.430334\ cm^{-1}$

Enthalpy of Formation

Drowart *et al.*[1] have measured mass spectrometrically the ion intensity ratio C_3^+/C^+ as a function of temperature from 1800 to 2700 K. We derive from a 2nd law analysis of their data $\Delta_f H°(2200\ K) = 185.2 \pm 3.3$ kcal·mol^{-1}, and $S°(2200\ K) = 74.9 \pm 2.8$ cal·K^{-1}·mol^{-1} by calculation of the absolute pressure from the vapor pressure of monatomic carbon and a relative ionization cross-section ratio of 2.3. The entropy uncertainty includes 1.3 cal·K^{-1}·mol^{-1} for uncertainty in the absolute pressure. Several earlier mass-spectrometric determinations[2-4] are in reasonable agreement. The absolute magnitude of the pressure is also confirmed by the work of Thorn and Winslow[5] who measured the total pressure of all species over graphite as 1.2 to 4.8×10^{-7} atm at 2400 K. Subtracting the JANAF values for C and C_2 we obtain $C_3 = 5.4 \times 10^{-8}$ to 2.2×10^{-7} atm, which compares with Drowart's value of 3.3×10^{-8} to 1.3×10^{-7} atm. The 2nd law enthalpy of formation is reported[5] to be $\Delta_f H°(2400\ K) = 184.4$ kcal·mol^{-1} but the range is only 80 K. It does, however, confirm the magnitude of the mass-spectrometric values for both enthalpy and entropy. We have adopted a enthalpy of formation within the uncertainty limits of Dowart's measurements which with the adopted functions yields pressures in the 2000–3000 K range consistent with the measurements. We were not able to generate functions which agree exactly with the 2nd law entropies, since even direct summation over only the observed levels of the bending frequency gives an entropy which is already too large.

Heat Capacity and Entropy

The stretching frequencies are from Merer,[6] Weltner *et al.*[7] and Weltner and McLeod.[8] The bending frequency and its harmonics are from Gausset *et al.*[9] and Merer.[6] The rotational constant and bond distance are from Gausset *et al.*[9]

As indicated briefly above, the calculated entropy of C_3 gas is higher than the experimental value even when a summation is performed only over the observed first six levels of the bending frequency. Treatment of the vibration as a harmonic oscillator or as a anharmonic oscillator gives values considerably higher. Strauss and Thiele[10] have also calculated functions based on a quartic potential function whlich still yields values several units higher than the experiments. In order to reduce the entropy to the approximate range of the measurements we have made the assumption that the potential function is of an unusual form above the sixth vibrational level. We assume that the large amplitude of the bending vibration causes a drastic change in the function, possibly due to the increasing repulsion between end carbons, as the amplitude increases. Thus, we assume a very steeply rising potential above the sixth vibrational level. We have chosen to represent this part of the potential by a harmonic oscillator of 650 cm^{-1}. The choice of 650 cm^{-1} is arbitrary, but is not unreasonable for a bending vibration, and yields functions which approximate the experimental data.

In order to explain the observed spectrum of C_3O_2, a molecule with a bending frequency of 63 cm^{-1}, Smith and Barrett[11] have proposed that potential maxima occur in the bending potential away from linearity. Thus, the adoption of an unusual potential function for C_3 bending may be justifiable. We do not claim that our functions are necessarily correct but only that they include all known facts and satisfy the experimental equilibrium observations. Further investigation of the carbon vapor pressure and of the higher vibrational levels of C_3 is desirable.

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Enthalpy Reference Temperature = T_r = 298.15 K Standard State Pressure = $p°$ = 0.1 MPa

T/K	$C_p°$	$S°$	$-[G°-H°(T_r)]/T$	$H°-H°(T_r)$	$\Delta_f H°$	$\Delta_f G°$	log K_f
	J·K^{-1}·mol^{-1}			kJ·mol^{-1}			
0	0.	0.	INFINITE	−11.761	811.456	811.456	INFINITE
100	42.459	193.021	272.308	−7.929	815.110	796.093	−415.836
200	39.861	221.795	240.748	−3.791	818.268	775.758	−202.607
298.15	37.740	237.246	237.246	0.	820.064	754.464	−132.179
300	37.719	237.479	237.247	0.070	820.086	754.057	−131.293
400	37.497	248.274	238.720	3.821	820.770	731.916	−95.578
500	38.380	256.727	241.504	7.612	820.580	709.709	−74.143
600	39.706	263.841	244.649	11.515	819.749	687.604	−59.861
700	41.125	270.069	247.844	15.557	818.472	665.676	−49.673
800	42.480	275.650	250.977	19.738	816.891	643.953	−42.046
900	43.714	280.726	254.005	24.049	815.098	622.441	−36.126
1000	44.815	285.390	256.913	28.477	813.156	601.138	−31.400
1100	45.786	289.708	259.701	33.008	811.105	580.035	−27.544
1200	46.647	293.729	262.371	37.630	808.974	559.122	−24.338
1300	47.413	297.494	264.929	42.334	806.781	538.389	−21.633
1400	48.095	301.033	267.383	47.110	804.538	517.828	−19.320
1500	48.710	304.372	269.739	51.951	802.255	497.428	−17.322
1600	49.267	307.534	272.003	56.850	799.940	477.181	−15.578
1700	49.773	310.536	274.182	61.802	797.595	457.081	−14.044
1800	50.233	313.394	276.282	66.803	795.226	437.119	−12.685
1900	50.660	316.122	278.307	71.848	792.834	417.289	−11.472
2000	51.053	318.731	280.264	76.934	790.421	397.586	−10.384
2100	51.417	321.230	282.155	82.057	787.989	378.004	−9.402
2200	51.760	323.630	283.986	87.216	785.538	358.538	−8.513
2300	52.081	325.938	285.761	92.408	783.069	339.184	−7.703
2400	52.384	328.161	287.481	97.632	780.584	319.938	−6.963
2500	52.667	330.305	289.152	102.885	778.081	300.796	−6.285
2600	52.936	332.376	290.775	108.165	775.562	281.754	−5.661
2700	53.189	334.379	292.353	113.471	773.025	262.809	−5.084
2800	53.430	336.318	293.888	118.802	770.472	243.959	−4.551
2900	53.659	338.197	295.384	124.157	767.902	225.200	−4.056
3000	53.877	340.019	296.842	129.534	765.315	206.531	−3.596
3100	54.084	341.789	298.263	134.932	762.711	187.947	−3.167
3200	54.283	343.510	299.650	140.350	760.089	169.448	−2.766
3300	54.476	345.183	301.005	145.788	757.450	151.031	−2.391
3400	54.660	346.812	302.328	151.245	754.793	132.695	−2.039
3500	54.834	348.399	303.622	156.720	752.118	114.437	−1.708
3600	55.003	349.946	304.887	162.212	749.425	96.256	−1.397
3700	55.164	351.455	306.126	167.720	746.713	78.149	−1.103
3800	55.321	352.929	307.338	173.244	743.982	60.116	−0.826
3900	55.472	354.368	308.526	178.784	741.233	42.156	−0.565
4000	55.618	355.774	309.689	184.339	738.464	24.266	−0.317
4100	55.757	357.149	310.830	189.907	735.676	6.445	−0.082
4200	55.893	358.494	311.949	195.490	732.868	−11.307	0.141
4300	56.024	359.811	313.047	201.086	730.040	−28.992	0.352
4400	56.151	361.100	314.124	206.695	727.192	−46.611	0.553
4500	56.275	362.364	315.182	212.316	724.324	−64.165	0.745
4600	56.392	363.602	316.222	217.949	721.434	−81.655	0.927
4700	56.506	364.816	317.243	223.594	718.524	−99.082	1.101
4800	56.618	366.007	318.246	229.250	715.593	−116.447	1.267
4900	56.726	367.175	319.233	234.918	712.641	−133.751	1.426
5000	56.831	368.322	320.203	240.596	709.667	−150.994	1.577
5100	56.932	369.449	321.158	246.284	706.672	−168.178	1.722
5200	57.031	370.555	322.097	251.982	703.654	−185.302	1.861
5300	57 127	371.642	323.022	257.690	700.614	−202.368	1.994
5400	57.221	372.711	323.932	263.407	697.552	−219.377	2.122
5500	57.312	373.762	324.828	269.134	694.468	−236.329	2.244
5600	57.402	374.795	325.712	274.870	691.361	−253.225	2.362
5700	57.488	375.812	326.582	280.614	688.232	−270.064	2.475
5800	57.573	376.813	327.439	286.367	685.080	−286.849	2.583
5900	57.655	377 798	328.284	292.129	681.904	−303.579	2.688
6000	57.735	378.767	329.118	297.898	678.706	−320.256	2.788

PREVIOUS December 1969 (1 atm) CURRENT: December 1969 (1 bar)